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New insights into preventing injuries and illnesses among elite athletics athletes

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For the athletics athlete, 'injury' is part of life; either injury and its potential consequences are in his/her mind like the sword of Damocles or injury is present. About two-thirds of athletes will sustain at least one injury during an athletics season.[1] Furthermore, it is not uncommon that athletes continue to train despite complaining of pain or injury.[2] But this should no longer be considered as the norm! Indeed, the normal condition of athletics practice should be to do so in the absence of any pain or injury. Efforts should be continued to reach this 'state of peace' in sports practice. Athletes may also suffer from illnesses, which will also have repercussions on their athletics training and performance.[3 4] While medical teams battle to keep their athletes fit and healthy, this is also of major importance to all those who have a stake in the athlete's performance. To move forward towards the common goal of prevention rather than cure, the highly relevant and helpful new findings are summarised below.[5–7]

HEALTH STATUS BEFORE A CHAMPIONSHIP IS ONE OF THE KEYS

About one-third of the athletes involved in an international athletics championship reported an injury complaint during the month before the championship.[2 6 8] This is a relatively high percentage in this population of high-level and highly trained athletes who would be expected to have good and/or optimal medical care.

Moreover, athletes who reported an injury complaint during the month before the championship had a twofold to fourfold risk of sustaining a new injury during the period of the championship.[2 6] Recently, Timpka *et al*[6] reported that illness symptoms causing anxiety in the month before championships were associated with a fivefold increase in the likelihood of in-championship injury. This relationship between health status before and during championships can be explained by both direct and indirect effects. Either the healing process is incomplete and tissue is weaker than needed for such high-level activity, exposing the athlete to the risk of reinjury, or physical conditioning and/or technical preparation have not been accomplished, increasing the risk of injury. Of course, the whole season itself also influences the health status in the month before and during the championship.

Thus, the health status during the last month preceding a championship seems to represent one of the keys for performing a healthy championship. In practice, we suggest carefully following athletes during this critical period by improving communication between athletes and medical teams and to screen the athletes' health status before championships, using a preparticipation health questionnaire, for example.[2 6 8]

PAY ATTENTION TO CLIMATIC CONDITIONS

The climatic conditions during international championships and athletics training can affect sport performance and also influence the risk of injury and illness. Since international championships often take place during the summer, athletes may have to exercise in the heat, exposing the athlete to an increased risk of heat illness.[4 6 7]

Among athletes participating at the World Athletics Championships, Périard *et al*[7] recently reported that approximately half of the athletes participating in the study had previously experienced heat illness symptoms (eg, cramping, severe headaches). However, only 15% of them reported having prepared specifically for the championships by undertaking heat training regimens.[7]

As a practical prevention measure, they suggest that information regarding heat acclimatisation should be provided to athletes and their staffs in order to protect the athletes' health and optimise performance during championships in heat conditions.[7]

PAY ATTENTION TO ENDURANCE ATHLETES

Endurance events are associated with a high risk of injury and illness during international championships.[1 4] Timpka *et al*[6] also recently reported that endurance athletes have a 10-fold increase in the likelihood of sustaining an illness during athletics championship. It is therefore important to focus on endurance athletes as a relevant athletics health protection strategy. Practical tips include: strict follow-up, screening of athletes at risk, education regarding basic hygiene and paying attention to climatic conditions.[4]

PREVENT THE MOST COMMON INJURY: HAMSTRING INJURY

Hamstring injury is the most frequent injury diagnosis in athletics during international championships and throughout the entire athletics season.[1 5] Parameters associated with a higher risk of hamstring injury during championships include being a man, increasing age and participating in explosive events,[5] which can help medical and technical staffs to better screen and/or monitor athletes at increased risk. Hamstring injury prevention should be sex specific with adapted physical and strengthening conditioning.[5]

CONTINUE ON THE TRACK OF INJURY AND ILLNESS PREVENTION!

These new insights will help improve and optimise health protection and injury prevention in athletes, both at the elite level and for more modest athletes. Efforts have been performed and must be continued to move forward into athletics injury and illness prevention![1]

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Contributors PE: substantial contributions to conception, drafting the article, critical revision of the article for important intellectual content and final approval of the version to be published. J-MA, PB: substantial contributions to critical revision of the article and final approval of the version to be published.

Competing interests None declared.

Provenance and peer review Not commissioned; externally peer reviewed.

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To cite Edouard P, Alonso J-M, Branco P. *Br J Sports Med* 2018;**52**:4–5.

Accepted 8 April 2017
Published Online First 5 June 2017

Br J Sports Med 2018;**52**:4–5.
doi:10.1136/bjsports-2016-097369

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Br J Sports Med 2018 52: 4-5 originally published online June 5, 2017
doi: 10.1136/bjsports-2016-097369

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